

Daily Algebra Drill #2 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Vieta's formulas for cubics · Nested surds · Telescoping products · Lagrange/Brahmagupta identity · Factor theorem · Cyclic sums under constraints.

Section A Rapid Recognition

(≤15 s each)

Factorise, simplify, or evaluate instantly. No expansion.

1. Factorise completely: $x^3 - 27$.

Answer: $(x - 3)(x^2 + 3x + 9)$

Method: Sum/difference of cubes: $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ with $a = x$, $b = 3$. The quadratic factor is irreducible over $\mathbb{R}$.

Investigate further: Show that $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$ by expanding the right-hand side. Then generalise: for which values of n does $(a - b)$ divide $a^n - b^n$? (Answer: all positive integers n .) Prove it using the factor theorem.

2. Without expanding, evaluate $(\sqrt{5} + \sqrt{3})(\sqrt{5} - \sqrt{3})$.

Answer: 2

Method: Conjugate product: $(a + b)(a - b) = a^2 - b^2 = 5 - 3 = 2$. This is the core move for rationalising surds.

Investigate further: Evaluate $(\sqrt{n+1} + \sqrt{n})(\sqrt{n+1} - \sqrt{n})$ for any positive integer n . Use this to write $\frac{1}{\sqrt{n+1} + \sqrt{n}}$ in a simpler form, then telescope $\sum_{n=1}^{99} \frac{1}{\sqrt{n+1} + \sqrt{n}}$.

3. Without a calculator, evaluate $999^2 - 1$.

Answer: 998 000

Method: $999^2 - 1^2 = (999 + 1)(999 - 1) = 1000 \times 998 = 998\,000$. Treat every “near-square minus 1” as a difference of two squares.

Investigate further: Prove that every product of two integers whose average is an integer can be written as a perfect square minus a perfect square. (Hint: if $p + q = 2m$, write $pq = m^2 - \left(\frac{p-q}{2}\right)^2$.) This is the basis of Fermat's factorisation method.

4. Given $a - b = 4$ and $ab = 5$, find $a^2 + b^2$.

Answer: 26

Method: $(a - b)^2 = a^2 - 2ab + b^2 = 16$, so $a^2 + b^2 = 16 + 2(5) = 26$. Never solve for a and b individually.

Investigate further: Form the quadratic with roots a and b : you know $a - b = 4$ and $ab = 5$, so $a + b = \pm\sqrt{(a - b)^2 + 4ab} = \pm\sqrt{36} = \pm 6$. Find $a^3 - b^3$ and $a^4 + b^4$ without finding a or b .

5. Expand and simplify: $(x + y)^3 - x^3 - y^3$.

Answer: $3xy(x + y)$

Method: Expand $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$ then cancel x^3 and y^3 . Factor out $3xy$. Memorise: $(x + y)^3 - x^3 - y^3 = 3xy(x + y)$.

Investigate further: This is the key ingredient in the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$. Verify this identity by expanding, then use it to prove that if $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$.

6. Factorise completely: $3x^2 - 75$.

Answer: $3(x - 5)(x + 5)$

Method: Take out the common factor of 3 first, giving $3(x^2 - 25)$, then apply difference of two squares. Always extract scalars before factorising.

Investigate further: Generalise: find all positive integers k for which $kx^2 - k \cdot n^2$ factors into three factors over the integers. When does $ax^2 + bx + c$ have a common integer factor across all three coefficients?

7. Rationalise the denominator: $\frac{1}{\sqrt{3} + \sqrt{2}}$.

Answer: $\sqrt{3} - \sqrt{2}$

Method: Multiply top and bottom by the conjugate $(\sqrt{3} - \sqrt{2})$. Denominator becomes $3 - 2 = 1$. The result is exact and requires no further work.

Investigate further: Show that $\frac{1}{\sqrt{n} + \sqrt{n-1}} = \sqrt{n} - \sqrt{n-1}$. Hence find a closed form for $\sum_{n=1}^N \frac{1}{\sqrt{n} + \sqrt{n-1}}$.

8. Factorise: $x^2(x - 1) - (x - 1)$.

Answer: $(x - 1)(x - 1)(x + 1) = (x - 1)^2(x + 1)$

Method: $(x - 1)$ is a common factor: $(x - 1)(x^2 - 1) = (x - 1)(x - 1)(x + 1)$. Always look for a common binomial factor before anything else.

Investigate further: Use the factor theorem: verify that $x = 1$ is a double root of $x^3 - x^2 - x + 1$ (which equals $x^2(x - 1) - (x - 1)$) by showing $f(1) = 0$ and $f'(1) = 0$. What does a double root mean geometrically on the graph of f ?

9. The roots of $x^2 - 5x + 3 = 0$ are p and q . Find $p^2 + q^2$.

Answer: 19

Method: By Vieta: $p + q = 5$, $pq = 3$. Then $p^2 + q^2 = (p + q)^2 - 2pq = 25 - 6 = 19$. This is Vieta +

Newton identity: essential toolkit.

Investigate further: Vieta's formulas for a cubic. If α, β, γ are roots of $x^3 + px^2 + qx + r = 0$, then $\alpha + \beta + \gamma = -p$, $\alpha\beta + \beta\gamma + \gamma\alpha = q$, $\alpha\beta\gamma = -r$. Use these to find $\alpha^2 + \beta^2 + \gamma^2$ in terms of p and q . (Answer: $p^2 - 2q$.) Verify with the cubic $x^3 - 6x^2 + 11x - 6 = 0$.

10. Simplify: $\frac{2^{n+3} - 2^n}{2^n}$.

Answer: 7

Method: Factor 2^n from the numerator: $\frac{2^n(2^3 - 1)}{2^n} = 8 - 1 = 7$. Powers of 2 always simplify by factoring out the smallest power.

Investigate further: Generalise: simplify $\frac{k^{n+m} - k^n}{k^n}$ for any base k and exponent m . Then investigate $\frac{2^{2n+1} + 2^{2n-1}}{2^{2n}}$: can you spot the pattern for alternating-parity exponents?

Section B Manipulation Drills

(≤50 s each)

Fractions, rearranging, hidden factorisations. Elegance over brute force.

1. Simplify: $\frac{x}{x^2 - 1} + \frac{1}{x - 1}$.

Answer: $\frac{2x + 1}{(x - 1)(x + 1)}$

Method: Common denominator is $(x - 1)(x + 1)$. Numerator: $x + (x + 1) = 2x + 1$. Do not expand the denominator.

Investigate further: Now add a third term: simplify $\frac{x}{x^2 - 1} + \frac{1}{x - 1} - \frac{1}{x + 1}$. Then investigate: for which x is the original expression negative?

2. Factorise: $x^4 - x^2 - 12$.

Answer: $(x^2 - 4)(x^2 + 3) = (x - 2)(x + 2)(x^2 + 3)$

Method: Let $u = x^2$: $(u - 4)(u + 3) = 0$. Only $u = 4$ gives real solutions. The factor $(x^2 + 3)$ is irreducible over $\mathbb{R}$.

Investigate further: Find all real solutions of $x^4 - x^2 - 12 = 0$. Then modify the constant: for what values of c does $x^4 - x^2 + c = 0$ have (i) four real roots, (ii) two real roots, (iii) no real roots? (*Substitute $u = x^2$ and analyse the discriminant.*)

3. Make r the subject of $S = \frac{a}{1 - r}$.

Answer: $r = 1 - \frac{a}{S} = \frac{S - a}{S}$

Method: $S(1 - r) = a \Rightarrow 1 - r = a/S \Rightarrow r = 1 - a/S$. This is the geometric series sum formula — essential for TMUA Paper 1.

Investigate further: Geometric series connection. The formula $S = \frac{a}{1-r}$ holds for $|r| < 1$. Starting from $S_n = a \frac{1-r^{n+1}}{1-r}$, make r the subject in terms of S_n , a , n . Why is there no closed-form solution in general? (This links to polynomial root-finding and the Abel–Ruffini theorem.)

4. Simplify: $\left(1 + \frac{1}{x}\right)\left(1 - \frac{1}{x+1}\right)$.

Answer: $\frac{x+1}{x} \cdot \frac{x}{x+1} = 1$

Method: Write each factor as a single fraction: $\frac{x+1}{x} \cdot \frac{x}{x+1}$. The product telescopes to 1. Recognise the cancellation *before* multiplying out.

Investigate further: Generalise: show that $\prod_{k=1}^n \left(1 + \frac{1}{k}\right)\left(1 - \frac{1}{k+1}\right)$ telescopes. Then evaluate $\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right)$ by factoring each term as $\frac{(k-1)(k+1)}{k^2}$ and telescoping.

5. Given $x + y = 3$ and $xy = 1$, find $x^3 + y^3$.

Answer: 18

Method: $x^3 + y^3 = (x + y)(x^2 - xy + y^2) = (x + y)[(x + y)^2 - 3xy] = 3(9 - 3) = 18$. Build up from symmetric functions — never solve for x and y .

Investigate further: Find $x^4 + y^4$ and $x^5 + y^5$ using the same data ($x + y = 3$, $xy = 1$). (*Key: Newton's identity* $p_n = (x + y)p_{n-1} - xy \cdot p_{n-2}$ gives a recurrence $p_n = 3p_{n-1} - p_{n-2}$.) Compute $p_1, p_2, \dots, p_6$ and spot any patterns modulo small integers.

6. Simplify: $\frac{(a+b)^2 - (a-b)^2}{4ab}$.

Answer: 1

Method: $(a+b)^2 - (a-b)^2 = 4ab$ by the identity $(P+Q)^2 - (P-Q)^2 = 4PQ$. The expression equals $4ab/4ab = 1$.

Investigate further: The identity $(a+b)^2 - (a-b)^2 = 4ab$ can be rewritten as $\left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2 = ab$, which says AM² – HM-related term² = ab . Use this to prove the AM–GM inequality $\frac{a+b}{2} \geq \sqrt{ab}$ for $a, b > 0$ by observing that a square is always ≥ 0 .

7. Factorise: $x^3 + 3x^2 - x - 3$.

Answer: $(x-1)(x+1)(x+3)$

Method: Group: $(x^3 + 3x^2) - (x + 3) = x^2(x+3) - (x+3) = (x+3)(x^2 - 1) = (x+3)(x-1)(x+1)$. Grouping is faster than the factor theorem here.

Investigate further: Use the factor theorem: check $x = 1, -1, -3$ in turn. Then compare the grouping method versus synthetic division — which is faster? Generalise: $x^3 + ax^2 - x - a = (x^2 - 1)(x + a)$ for any a . For which cubics does grouping always work?

8. Solve: $\frac{x-1}{x+2} = \frac{x+3}{x+8}$.

Answer: $x = 7$

Method: Cross-multiply: $(x-1)(x+8) = (x+3)(x+2)$. Expand both sides: $x^2 + 7x - 8 = x^2 + 5x + 6$. The x^2 terms cancel: $2x = 14$, so $x = 7$. Check: LHS = $\frac{6}{9} = \frac{2}{3}$, RHS = $\frac{10}{15} = \frac{2}{3}$.

Investigate further: Notice the x^2 terms always cancel when both sides are linear-over-linear with the same degree numerator and denominator. Investigate: for which values of a, b, c, d does $\frac{x+a}{x+b} = \frac{x+c}{x+d}$ have no solutions? Exactly one? Infinitely many?

9. Simplify: $\frac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}} + \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}$.

Answer: $\frac{2(a+b)}{a-b}$

Method: Let $P = \sqrt{a} + \sqrt{b}$, $Q = \sqrt{a} - \sqrt{b}$. The expression is $\frac{Q}{P} + \frac{P}{Q} = \frac{P^2+Q^2}{PQ}$. $P^2 + Q^2 = 2(a+b)$, $PQ = a-b$.

Investigate further: Compare this with the structure $t + \frac{1}{t}$ where $t = \frac{\sqrt{a}-\sqrt{b}}{\sqrt{a}+\sqrt{b}}$. Note $t \in (-1, 1)$ for $a, b > 0$. Show that $t + 1/t \geq 2$ is false when $|t| < 1$ — this is why AM–GM requires *positive* quantities. What is the minimum of $\frac{2(a+b)}{a-b}$ subject to $a > b > 0$?

10. The cubic $x^3 - 6x^2 + 11x - 6 = 0$ has roots α, β, γ . Without solving it, find $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha\beta\gamma$.

Answer: $\alpha^2 + \beta^2 + \gamma^2 = 14$; $\alpha\beta\gamma = 6$

Method: By Vieta's formulas: $\alpha + \beta + \gamma = 6$, $\alpha\beta + \beta\gamma + \gamma\alpha = 11$, $\alpha\beta\gamma = 6$. Then $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = 36 - 22 = 14$.

Investigate further: Newton's power sums for a cubic. Define $p_k = \alpha^k + \beta^k + \gamma^k$. You found $p_1 = 6$ and $p_2 = 14$. Use Newton's identity: $p_k = (\alpha + \beta + \gamma)p_{k-1} - (\alpha\beta + \beta\gamma + \gamma\alpha)p_{k-2} + \alpha\beta\gamma \cdot p_{k-3}$ to find p_3 without solving the cubic. Verify by checking $\alpha = 1, \beta = 2, \gamma = 3$.

Section C Substitution & Structure

(≤ 90 s each)

Find structure before computing. Substitution is the key, not expansion.

1. Solve: $x^4 - 13x^2 + 36 = 0$.

Answer: $x = \pm 2, \pm 3$

Method: Let $u = x^2$: $(u-4)(u-9) = 0$, giving $u = 4$ or $u = 9$. Both positive, so four real solutions $x = \pm 2, \pm 3$.

Investigate further: Now solve $x^4 - 13x^2 + 36 = k$ for $k = 10, k = -1$. For which values of k does the equation have 4, 2, or 0 real solutions? Sketch $y = x^4 - 13x^2 + 36$ and identify the local minima and maxima algebraically (without calculus) by completing the square in $u = x^2$.

2. Solve: $(x+1)(x+2)(x+3)(x+4) = 24$.

Answer: $x = 0$ and $x = -5$

Method: Pair the outer and inner terms: $(x+1)(x+4) = x^2 + 5x + 4$ and $(x+2)(x+3) = x^2 + 5x + 6$. Let $u = x^2 + 5x + 5$: $(u-1)(u+1) = 24 \Rightarrow u^2 = 25 \Rightarrow u = \pm 5$. $u = 5$: $x^2 + 5x = 0 \Rightarrow x(x+5) = 0$. $u = -5$: $x^2 + 5x + 10 = 0$, discriminant < 0 , no real solutions.

Investigate further: This “pair the ends” trick works whenever the four roots are in arithmetic progression. Generalise: solve $(x+a)(x+b)(x+c)(x+d) = k$ where $a+d = b+c$. Then try: $(x)(x+2)(x+4)(x+6) = 9$.

3. Simplify: $\sqrt{6 + 2\sqrt{5}}$.

Answer: $\sqrt{5} + 1$

Method: Write $6 + 2\sqrt{5} = 5 + 2\sqrt{5} + 1 = (\sqrt{5} + 1)^2$. Then $\sqrt{(\sqrt{5} + 1)^2} = \sqrt{5} + 1$ (positive root).

Investigate further: Denesting nested surds. To simplify $\sqrt{a + b\sqrt{c}}$, guess the form $\sqrt{p} + \sqrt{q}$ and solve $p + q = a$, $4pq = b^2c$. Use this method to simplify $\sqrt{7 + 2\sqrt{10}}$ and $\sqrt{11 - 4\sqrt{7}}$. When does $\sqrt{a + b\sqrt{c}}$ denest to a rational combination of simple surds?

4. Given $a + b + c = 1$ and $a^2 + b^2 + c^2 = 1$, find $ab + bc + ca$ and hence $a^3 + b^3 + c^3 - 3abc$.

Answer: $ab + bc + ca = 0$; $a^3 + b^3 + c^3 - 3abc = 1$

Method: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca) \Rightarrow 1 = 1 + 2(ab + bc + ca)$, so $ab + bc + ca = 0$. Then apply the factorisation $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 1 \cdot (1 - 0) = 1$. Note abc itself is *not* determined by the given data — only the combination $a^3 + b^3 + c^3 - 3abc$ is.

Investigate further: Use Newton’s identity $p_3 = e_1p_2 - e_2p_1 + 3e_3$ (where e_1, e_2, e_3 are the elementary symmetric polynomials) to find $a^3 + b^3 + c^3$ in terms of e_1, e_2, e_3 in general. Verify with $a = 1, b = 0, c = 0$. Then investigate: if $a^n + b^n + c^n$ is an integer for $n = 1, 2, 3$, must it be an integer for all n ? (Ramanujan connection.)

5. Solve: $(x-1)(x-3)(x-5)(x-7) + 15 = 0$.

Answer: $x = 2, 6, 4 \pm \sqrt{6}$ (all four real)

Method: Pair: $(x-1)(x-7) = x^2 - 8x + 7$ and $(x-3)(x-5) = x^2 - 8x + 15$. $(u-4)(u+4) = u^2 - 16$, so $u^2 - 16 + 15 = u^2 - 1 = 0 \Rightarrow u = \pm 1$. $u = 1$: $x^2 - 8x + 10 = 0 \Rightarrow x = 4 \pm \sqrt{6}$. $u = -1$: $x^2 - 8x + 12 = 0 \Rightarrow (x-2)(x-6) = 0$.

Investigate further: The “pair the ends” substitution $u = x^2 - 8x + \text{const}$ works because the four linear factors are symmetric about $x = 4$ (the mean of 1, 3, 5, 7). Investigate: $(x-a)(x-b)(x-c)(x-d) + k = 0$ where $a+d = b+c$. Show that the substitution $u = x^2 - (a+d)x + \frac{(a+d)^2}{4}$ always reduces this to a quadratic in u .

6. If $x = 2 + \sqrt{3}$, find $x^2 + \frac{1}{x^2}$ without expanding x^2 directly.

Answer: 14

Method: Note $\frac{1}{x} = \frac{1}{2+\sqrt{3}} = 2 - \sqrt{3}$ (conjugate rationalisation). So $x + \frac{1}{x} = (2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$. Then $x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2 = 16 - 2 = 14$.

Investigate further: Observe that $x = 2 + \sqrt{3}$ and $\frac{1}{x} = 2 - \sqrt{3}$ are conjugate surds. Show that for $x = a + \sqrt{b}$ (with a, b rational), $\frac{1}{x} = \frac{a - \sqrt{b}}{a^2 - b}$. When does $x + \frac{1}{x}$ become an integer? Find all integers n such that $x + \frac{1}{x} = n$ for some $x = a + \sqrt{b}$ with $a, b \in \mathbb{Z}$.

7. Given $\frac{a}{b} + \frac{b}{a} = 3$, find $\frac{a^2}{b^2} + \frac{b^2}{a^2}$.

Answer: 7

Method: Let $t = \frac{a}{b} + \frac{b}{a} = 3$. Then $\frac{a^2}{b^2} + \frac{b^2}{a^2} = \left(\frac{a}{b} + \frac{b}{a}\right)^2 - 2 = 9 - 2 = 7$.

Investigate further: Extend the chain: find $\frac{a^3}{b^3} + \frac{b^3}{a^3}$ using the identity $t^3 = \left(\frac{a}{b}\right)^3 + \left(\frac{b}{a}\right)^3 + 3 \cdot \frac{a}{b} \cdot \frac{b}{a} \left(\frac{a}{b} + \frac{b}{a}\right)$. Generalise: if $s_n = \left(\frac{a}{b}\right)^n + \left(\frac{b}{a}\right)^n$, show that $s_n = 3s_{n-1} - s_{n-2}$ and compute s_1, s_2, s_3, s_4, s_5 .

8. If $x + y + z = 0$, simplify $\frac{(x^2 + y^2 + z^2)^2}{x^4 + y^4 + z^4}$.

Answer: 2

Method: Let $s = x^2 + y^2 + z^2$. Since $x + y + z = 0$: $(x + y + z)^2 = 0 \Rightarrow x^2 + y^2 + z^2 = -2(xy + yz + zx)$, so $xy + yz + zx = -s/2$. Then $(x^2 + y^2 + z^2)^2 = x^4 + y^4 + z^4 + 2(x^2y^2 + y^2z^2 + z^2x^2)$. Also $(xy + yz + zx)^2 = x^2y^2 + y^2z^2 + z^2x^2 + 2xyz(x + y + z) = x^2y^2 + y^2z^2 + z^2x^2$. So $s^2 = x^4 + y^4 + z^4 + 2 \cdot s^2/4$, giving $x^4 + y^4 + z^4 = s^2/2$. The ratio is $s^2/(s^2/2) = 2$.

Investigate further: This result says that under the constraint $x + y + z = 0$, the ratio of the square of the second power sum to the fourth power sum is always 2. Investigate the analogous ratio $\frac{(x^2 + y^2 + z^2)^3}{x^6 + y^6 + z^6}$ when $x + y + z = 0$. Also: prove that $x^4 + y^4 + z^4 = \frac{1}{2}(x^2 + y^2 + z^2)^2$ is equivalent to $(xy + yz + zx)^2 = 2xyz(x + y + z)$, which is automatic when the sum is zero.

Section D Challenge

(TMUA / SMC difficulty)

Each has a solution under five lines. Find the insight, not the computation.

1. Evaluate:

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{3 \cdot 4 \cdot 5} + \cdots + \frac{1}{8 \cdot 9 \cdot 10}.$$

Answer: $\frac{11}{45}$

Method: Partial fractions: $\frac{1}{n(n+1)(n+2)} = \frac{1}{2} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right)$. This is a telescoping sum! Sum from $n = 1$ to 8: $\frac{1}{2} \left(\frac{1}{1 \cdot 2} - \frac{1}{9 \cdot 10} \right) = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{90} \right) = \frac{1}{2} \cdot \frac{44}{90} = \frac{22}{90} = \frac{11}{45}$.

Investigate further: Higher-order telescoping. Generalise the partial fraction: $\frac{1}{n(n+1) \cdots (n+k)} = \frac{1}{k} \left(\frac{1}{n(n+1) \cdots (n+k-1)} - \frac{1}{(n+1) \cdots (n+k)} \right)$. Use this to find a closed form for $\sum_{n=1}^N \frac{1}{n(n+1)(n+2)(n+3)}$ and verify for $N = 2$.

2. The roots of $x^2 + px + q = 0$ are r and s . Find, in terms of p and q , the monic quadratic whose

roots are $r^2 + s$ and $s^2 + r$.

Answer: $x^2 - (p^2 - 2q - p)x + (q^2 - p^3 + 3pq + q) = 0$

Method: By Vieta on the original: $r + s = -p$, $rs = q$. New sum: $(r^2 + s) + (s^2 + r) = (r^2 + s^2) + (r + s) = [(r + s)^2 - 2rs] + (r + s) = (p^2 - 2q) + (-p) = p^2 - 2q - p$. New product: $(r^2 + s)(s^2 + r) = r^2s^2 + r^3 + s^3 + rs = q^2 + (r + s)^3 - 3rs(r + s) + q = q^2 + (-p)^3 - 3q(-p) + q = -p^3 + 3pq + q^2 + q$. Required quadratic: $x^2 - (p^2 - 2q - p)x + (-p^3 + 3pq + q^2 + q) = 0$.

Investigate further: This is a classic example of constructing a new polynomial from transformed roots — a technique called *Tschirnhaus transformation*. Try the simpler version: find the quadratic with roots r^2 and s^2 . Then find the quartic with roots r^2, s^2, r^3, s^3 (this is much harder — think about what Vieta gives you for degree 4).

3. The equation $\frac{x}{y+z} = \frac{y}{x+z} = \frac{z}{x+y} = k$ holds for real x, y, z with $x + y + z \neq 0$. Find all possible values of k .

Answer: Under $x + y + z \neq 0$: $k = \frac{1}{2}$ only.

Method: Set each ratio equal to k : $x = k(y + z)$, $y = k(x + z)$, $z = k(x + y)$. Adding gives $x + y + z = 2k(x + y + z)$. Since $x + y + z \neq 0$, divide through to get $k = \frac{1}{2}$.

Investigate further: Investigate the case $x + y + z = 0$ separately: show that each ratio equals -1 . (When $x + y + z = 0$, we have $y + z = -x$, so $\frac{x}{y+z} = \frac{x}{-x} = -1$.) Hence in general, without any constraint, the possible values of k are $\frac{1}{2}$ and -1 . Find $(x : y : z)$ for each case.

4. Prove the **Brahmagupta–Fibonacci identity**:

$$(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2.$$

Hence express 85×65 as a sum of two perfect squares in two different ways.

Answer: $85 \times 65 = 5525 = 74^2 + 7^2 = 71^2 + 22^2$.

Method: *Proof:* Expand the right-hand side: $(ac + bd)^2 + (ad - bc)^2 = a^2c^2 + 2abcd + b^2d^2 + a^2d^2 - 2abcd + b^2c^2 = a^2(c^2 + d^2) + b^2(c^2 + d^2) = (a^2 + b^2)(c^2 + d^2)$. $\square$

Representations: Use $85 = 2^2 + 9^2$, and $65 = 1^2 + 8^2$ and $65 = 4^2 + 7^2$. $(2^2 + 9^2)(1^2 + 8^2) = (2 \cdot 1 + 9 \cdot 8)^2 + (2 \cdot 8 - 9 \cdot 1)^2 = 74^2 + 7^2 = 5476 + 49 = 5525$. $\checkmark$ $(2^2 + 9^2)(4^2 + 7^2) = (8 + 63)^2 + (14 - 36)^2 = 71^2 + 22^2 = 5041 + 484 = 5525$. $\checkmark$

Investigate further: The Brahmagupta–Fibonacci identity has a beautiful complex-number proof: $|z_1|^2 |z_2|^2 = |z_1 z_2|^2$ where $z_1 = a + bi$, $z_2 = c + di$. Expand $z_1 z_2$ and read off the identity. Use this to determine which positive integers can be expressed as a sum of two squares. (*Theorem: a positive integer n is a sum of two squares iff in its prime factorisation, every prime of the form $4k + 3$ appears to an even power.*)

5. Find all integers $n \geq 1$ for which $n^2 + n + 1$ divides $n^3 - 1$.

Answer: All integers $n \geq 1$.

Method: Use the identity $n^3 - 1 = (n - 1)(n^2 + n + 1)$. Therefore $n^2 + n + 1$ divides $n^3 - 1$ for every integer n , hence for all $n \geq 1$.

Investigate further: Polynomial divisibility. The factorisation $n^3 - 1 = (n - 1)(n^2 + n + 1)$ is a special case of the cyclotomic polynomial $\Phi_3(n) = n^2 + n + 1$. Investigate: for which n does $\Phi_3(n) = n^2 + n + 1$ itself have a factor in common with $n^2 - 1$? (*Answer:* $\gcd(n^2 + n + 1, n^2 - 1) = \gcd(n + 2, n^2 - 1)$, which divides 3.) Explore the analogous questions for $\Phi_4(n) = n^2 + 1$ and $\Phi_6(n) = n^2 - n + 1$.

Top 5 Patterns Today

- Vieta's formulas (quadratic and cubic).** $e_1 = \sum \alpha$, $e_2 = \sum \alpha\beta$, $e_3 = \alpha\beta\gamma$ read directly from the polynomial. Combine with Newton's power-sum identities to find $\alpha^k + \beta^k + \gamma^k$ without solving.
- Denesting nested surds.** $\sqrt{a + b\sqrt{c}} = \sqrt{p} + \sqrt{q}$ where $p + q = a$ and $4pq = b^2c$. Always try to write the radicand as a perfect square.
- Telescoping products and higher-order partial fractions.** $\frac{1}{n(n+1)(n+2)} = \frac{1}{2} \left(\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \right)$. Every product of consecutive integers telescopes with the right partial fraction.
- Brahmagupta–Fibonacci / complex modulus.** $(a^2 + b^2)(c^2 + d^2) = (ac \pm bd)^2 + (ad \mp bc)^2$. Products of sums of two squares are themselves sums of two squares.
- Constraint exploitation and cyclic sum collapse.** When $x + y + z = 0$, ratios of power sums become constants. When $a + b + c = 0$: use $a^3 + b^3 + c^3 = 3abc$; when $abc = 1$: multiply fractions by abc .

Common Traps to Avoid

- Expanding instead of pairing.** In C2 and C5, expanding all four brackets is fatal. Always look for symmetric pairs that produce the same sub-expression.
- Forgetting the second Vieta formula.** $\alpha\beta + \beta\gamma + \gamma\alpha$ (not $\alpha\beta\gamma$) gives the coefficient of x in a monic cubic. Confusing e_2 and e_3 is the most common Vieta error.
- Taking the wrong square root in nested surds.** $\sqrt{11 - 4\sqrt{7}} = 2 - \sqrt{7}$ only if $2 - \sqrt{7} > 0$, i.e. $\sqrt{7} < 2$, which is false. The answer is $\sqrt{7} - 2$. Always check positivity.
- Applying AM–GM to non-positive quantities.** AM–GM ($\frac{a+b}{2} \geq \sqrt{ab}$) requires $a, b \geq 0$. Applying it when quantities can be negative gives false inequalities.
- Assuming $\frac{x}{y+z} = \frac{y}{x+z} = \frac{z}{x+y}$ forces $x = y = z$.** The case $x + y + z = 0$ (giving $k = -1$) is equally valid and must not be overlooked.

“An elegant solution is not luck — it is the reward for seeing the structure that was always there.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths